

Subinhibitory concentration stress of colistin enhanced PhoPQ expression in *Escherichia coli* harboring *mcr-1*

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Abstract

The increased and inappropriate use of colistin led to the emergence of its resistance among Gram-negative bacterial isolates and the most common mechanism of colistin resistance in Gram-negative bacteria is the modification of the lipopolysaccharide mediated by two-component regulatory systems, PhoPQ and PmrAB. The aim of the present study was to investigate the transcriptional expression of the PhoPQ system against colistin stress in clinical isolates of *Escherichia coli* with colistin-resistant phenotype. Six colistin-resistant *E. coli* isolates were obtained from Silchar Medical College and Hospital, Silchar that were of clinical origin and received for routine culture and sensitivity testing. Screening for colistin resistance was done by broth microdilution method and further screened for the presence of the different types of plasmid-mediated colistin resistance *mcr* genes namely, *mcr-1* to *mcr-10* by polymerase chain reaction (PCR). The screened positive isolates were subjected to PCR assay targeting *phoP* and *phoQ* genes and their expression was measured by quantitative real-time PCR. The results of this study revealed that two *E. coli* isolates (TS2 and TS4) were found to carry the *mcr-1* gene. *PhoP* and *PhoQ* gene amplification was observed in all the isolates. Transcriptional analysis showed that the isolates harboring the *mcr-1* gene showed an enhanced level of expression in the *PhoP*, *PhoQ* genes in the presence of a subinhibitory concentration of colistin whereas no significant expression was observed for the isolates which were devoid of the *mcr* gene. This study demonstrates the involvement of *mcr-1* in the PhoPQ system in clinical isolates of colistin-resistant *E. coli* which will help in designing a molecular marker for detecting colistin-resistant *E. coli* and contribute to the assessment of resistance burden and infection control strategy.

KEYWORDS

PhoPQ, colistin-resistant, *Escherichia coli*, *mcr-1*, polymerase chain reaction

1 | INTRODUCTION

Colistin (polymyxin E) is a polycationic antimicrobial agent which is considered as a drug of last resort for the treatment of multidrug-resistant (MDR) Gram-negative bacteria in human medicine [1]. The increased and inappropriate use of this antibiotic led to the emergence of its resistance among Gram-negative bacterial isolates which has been frequently reported [2]. The most common mechanism of colistin resistance in Gram-negative bacteria is the modification of the lipid A component of the outer membrane lipopolysaccharide with phosphoethanolamine and 4-amino-4-deoxy-L-arabinose (L-Ara4N) mediated mainly by two-component regulatory systems, PhoPQ and PmrAB [3]. This modification neutralizes the negative charge of lipopolysaccharide thereby reducing the binding affinity of the colistin [4]. Overexpression and amino acid alterations of PhoPQ and PmrAB are known to be associated with resistance to colistin in various Gram-negative pathogens including Enterobacteriaceae, *Pseudomonas aeruginosa*, and *Acinetobacter baumannii* [5,6]. In the presence of cationic antimicrobial peptides (CAMPs), depletion of divalent cations such as Mg^{2+} and Ca^{2+} , or due to mutation, PhoPQ is activated [7,8]. Thus, it will be of immense interest to know how the PhoPQ system responds when colistin-resistant *Escherichia coli* are grown under sub-inhibitory concentration stress of colistin. Moreover, plasmid-mediated colistin resistance gene, *mcr*, capable of horizontal gene transfer has also been reported in different parts of the world [9]. They encode phosphoethanolamine transferase proteins that are responsible for the modification of the lipid A component of lipopolysaccharide leading to colistin resistance [10]. Several types of *mcr*, namely, *mcr-1*–*mcr-10* have been identified and reported in different countries [11,12].

In the current study, we have investigated the transcriptional response of the PhoPQ system against sub-inhibitory concentration colistin stress in clinical isolates of *E. coli* with colistin-resistant phenotype.

2 | MATERIALS AND METHODS

2.1 | Bacterial isolates

Six isolates of *E. coli* which were screened as potential colistin-resistant were collected from Silchar Medical College and Hospital, Silchar, from August 2019 to December 2019. The isolates were identified in the department of microbiology of the medical college from routine clinical samples (Table S1) processed for culture and sensitivity testing. Identification of the organisms

was performed with a VITEK® 2 compact system (Bio-merieux) and confirmed by 16S ribosomal RNA gene sequencing. Screening of the isolates for colistin resistance was done by the broth microdilution method using colistin (Hi-Media). The concentration of the colistin antibiotic used was 2–4 µg/ml and the results were interpreted as per EUCAST guidelines 2019 [12].

2.2 | Screening of plasmid-mediated colistin resistance

Polymerase chain reaction (PCR) assay was performed for screening of different types of plasmid-mediated colistin resistance genes namely, *mcr-1*, 2, 3, 4, 5, 6, 7, 8.2, 8.4, 9, and *mcr-10*. Specific oligonucleotides responsible for each gene were employed for the amplification (Table 1). Previously confirmed *mcr-1* positive isolate was used as a positive control.

2.3 | Genotypic characterization

The screened positive isolates were subjected to a PCR assay targeting *phoP* and *phoQ* genes. Specific oligonucleotide primers (Table 1) were used for the amplification of the genes. The PCR conditions were initial denaturation at 94°C for 3 min followed by 35 cycles of denaturation at 94°C for 40 s, annealing at 50°C for 30 s, extension at 72°C for 50 s and final extension at 72°C for 7 min.

2.4 | Transcriptional analysis of PhoPQ by quantitative real-time PCR

The expression of the *phoP* and *phoQ* genes was measured by quantitative real-time PCR (Applied Biosystem) using specific oligonucleotide primers (Table 1). Six colistin-resistant isolates were selected and *E. coli* ATCC 25922 was used as a reference strain. The isolates in addition to the control were inoculated in 5 ml Luria-Bertani broth (Hi-Media) without antibiotic pressure and also with colistin pressure (0.5 µg/ml of colistin). All the inoculated cultures were incubated in a shaker incubator (160 rpm) at 37°C for 12–16 h. Total messenger RNA (mRNA) was extracted from mid-log bacterial cultures (optical density at 600 nm) by using an RNeasy mini kit (Qiagen) following the manufacturer's instructions. Further, the mRNA was reverse-transcribed into complementary DNA by using the Quantiscript Reverse Transcription kit (Qiagen). Quantitative real-time PCR was performed using Power Sybr Green

TABLE 1 Oligonucleotides used in this study

Primer pairs	Target	Sequence (5'–3')	Amplified product size (bp)	Reference
Mcr A-F	<i>mcr-1</i>	CGCTTAAAATACGCAGGCC	521	[13]
Mcr B-R		GCAACCAAGCCTGATATGCG		
Mcr2F	<i>mcr-2</i>	CGCTTGGGTGTGGGTATCTT	323	This study
Mcr2R		TGGTGTTCGCATTGGCAAG		
Mcr3F	<i>mcr-3</i>	TATGCTGAAGTGGCGTGGAG	736	This study
Mcr3R		AAAGGAGATCACGCCACCAG		
Mcr4F	<i>mcr-4</i>	CTGGCAGTAGCACCAAGTCA	421	This study
Mcr4R		CTCGCTGTGCAACACAAGTC		
Mcr5F	<i>mcr-5</i>	TACGTCATCTCGGGCATTCTG	673	This study
Mcr5R		CGTAGGTGTTACCAAGGCT		
Mcr6F	<i>mcr-6</i>	AAGTATCTTGGTGGCGTGGG	844	This study
Mcr6R		ATTACCTTCGCACACAGGGG		
Mcr7F	<i>mcr-7</i>	AGCTGGTCGGTTTCCATCTG	250	This study
Mcr7R		TCGCCGTGATCAGAGAGGTA		
Mcr8.2F	<i>mcr-8.2</i>	AGCACCTCAACACCTGTACG	200	This study
Mcr8.2R		TATTCGCGGGTAACCAACCC		
Mcr8.4F	<i>mcr-8.4</i>	CCTGCATGTTCTCGGAATG	486	This study
Mcr8.4R		GCATCCCGGAATAACGTTGC		
Mcr9F	<i>mcr-9</i>	GTGTTTCCCGGCGATAAAGC	640	This study
Mcr9R		ATCCGTTCCGTGCATGTTCT		
Mcr10F	<i>mcr-10</i>	GGACCGACCTATTACCAGCG	366	[14]
Mcr10R		GGCATTATGCTGCAGACACG		
Phope1	PhoP	TGGTATGACTTTCCCGCAGC	118	This study
Phope2		ACCGCGTTCGAATACACCAT		
Phoqe1	PhoQ	ATGGTTGCGCTGATCGGTTA	101	This study
Phoqe2		TTTTCCCACTTCGCAAGGGT		

Master Mix (Applied Biosystem) using Step One Plus real-time detection system (Applied Biosystem). The transcript levels were normalized to the expression level of the housekeeping gene *rpsL* of *E. coli*. The fold-changes in the expression of *PhoP* and *PhoQ* in two different conditions were calculated according to the threshold cycle ($\Delta\Delta C_t$) method.

3 | RESULTS

Among six selected colistin-resistant *E. coli* isolates, two isolates, namely, TS2 and TS4 were found to carry the *mcr-1* gene, whereas the other four isolates were

devoid of the acquired colistin resistance gene. *PhoP* and *PhoQ* gene amplification was observed in all the isolates. While analyzing transcriptional expression, it was observed that *PhoP* or *PhoQ* did not show a significant increase in colistin-resistant isolates when compared with *E. coli* ATCC 25922 (Figures 1 and 2). While grown under the subinhibitory concentration of colistin stress, four isolates that are colistin-resistant and devoid of any *mcr* gene were not showing any significant difference in expression when compared with the isolates which were grown without antibiotic pressure. However, in isolates harboring the *mcr-1* gene, an enhanced level of expression upto ninefold (TS2) was observed in the case of the *PhoP* gene and

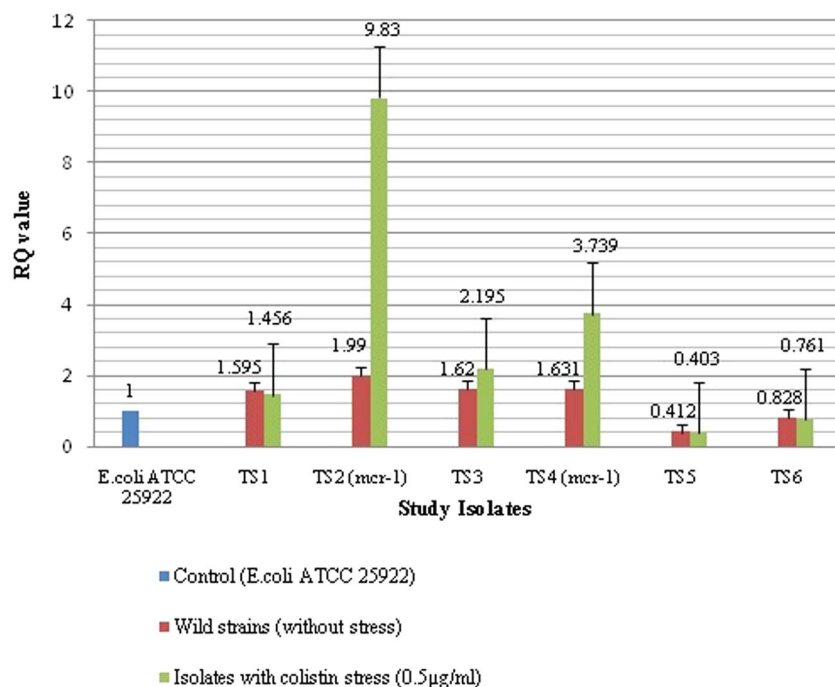


FIGURE 1 Expression of *PhoP* gene under normal condition (without stress) and colistin stress relative to *Escherichia coli* ATCC 25922

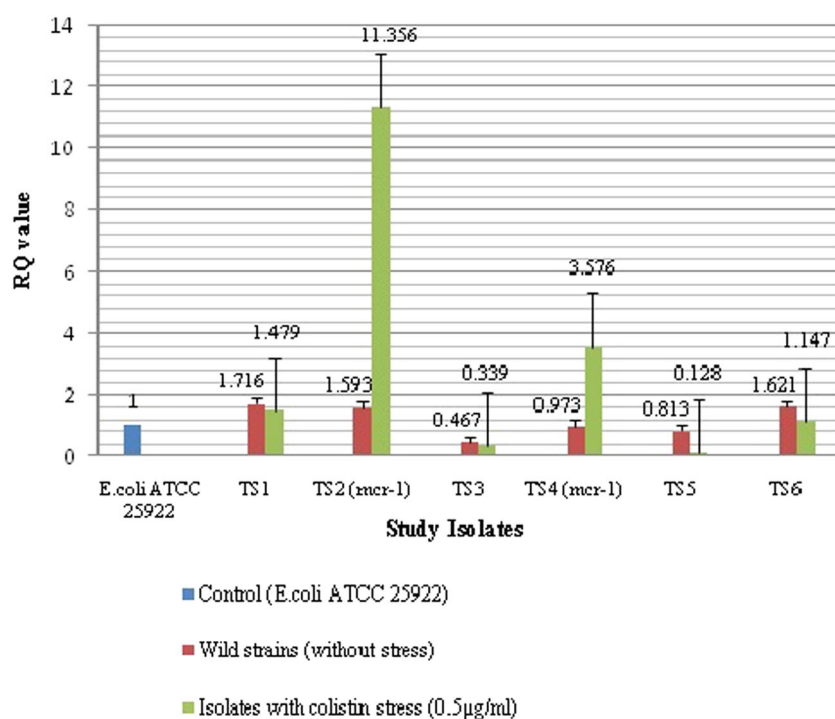


FIGURE 2 Expression of *PhoQ* gene under normal condition (without stress) and colistin stress relative to *Escherichia coli* ATCC 25922

upto 11 fold (TS2) for the *PhoQ* gene was observed (Figures 1 and 2).

4 | DISCUSSION

Colistin resistance in Gram-negative bacteria is associated with specific modification of the lipid A such as the addition of phosphoethanolamine or 4-amino-4-deoxy-L-

arabinose (L-Ara4N) mediated by the two-component regulatory systems such as PmrAB and PhoPQ. Mutations and overexpression of these two-component systems associated with colistin resistance in Gram-negative pathogens have been reported [5]. The current study focused on the PhoPQ component system which helps in regulating the expression of genes for lipid A modifications leading to colistin resistance in *E. coli*. The PhoPQ TCS encodes *PhoP* as a regulator protein and *PhoQ* as a sensor kinase in which

PhoQ activates *PhoP* by its kinase activity which activates transcription of the *pmrFHIIJKLM* operon, followed by modification of lipid A [2]. However, the impact of colistin resistance due to chromosomal mutations and gene expressional studies has been rarely investigated in *E. coli* [15]. Unlike previous studies [15,16], we did not observe any significant upregulation of *PhoP*, *PhoQ* genes (except TS2) when compared with *E. coli* ATCC 25922. A previous study has investigated the expression level of *PhoP*, *PhoQ* genes in clinical isolates of *E. coli* and found that the expression levels varied considerably and could not distinguish the colistin-resistant isolates from the colistin sensitive isolates [10]. Moreover, various other studies have mentioned that under conditions of low magnesium or calcium, acidic pH, or cationic antimicrobial peptide (colistin), *PhoPQ* is activated and protects Gram-negative bacteria such as *Salmonella* and *Pseudomonas* [17,18]. Here, we analyzed how *PhoPQ* responds to subinhibitory concentration colistin stress to induce colistin resistance in *E. coli*. In this study, we have observed that transcriptional expression of *PhoPQ* is enhanced in the presence of a subinhibitory concentration of colistin in the isolate when augmented with the *mcr-1* gene. The presence of *mcr-1* might be a factor for the enhanced expression of *PhoP*, *PhoQ* genes in *E. coli* isolates in the presence of colistin thereby increasing its resistance towards this particular antibiotic. This warrants further investigation to understand the possible genetic interplay between the *PhoPQ* system and the *mcr-1* gene.

This study demonstrates the involvement of *mcr-1* in the *PhoPQ* system in clinical isolates of colistin-resistant *E. coli* which will help in designing a molecular marker for detecting colistin-resistant *E. coli* and contribute to the assessment of resistance burden and infection control strategy.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

- [1] Kim S, Woo JH, Kim N, Kim MH, Kim SY, Son JH, et al. Characterization of chromosome-mediated colistin resistance in *Escherichia coli* isolates from livestock in Korea. *Infect Drug Resist.* 2019;12:3291–3299.
- [2] Aghapour Z, Gholizadeh P, Ganbarov K, Bialvaei AZ, Mahmood SS, Tanomand A, et al. Molecular mechanisms related to colistin resistance in Enterobacteriaceae. *Infect Drug Resist.* 2019;12:965–75.
- [3] Olaitan AO, Morand S, Rolain JM. Mechanisms of polymyxin resistance: acquired and intrinsic resistance in bacteria. *Front Microbiol.* 2014;5:643.
- [4] Andrade FF, Silva D, Rodrigues A, Pina-Vaz C. Colistin update on its mechanism of action and resistance, present and future challenges. *Microorganisms.* 2020;8:1716.
- [5] Hong JK, Ko KS. *PmrAB* and *PhoPQ* variants in colistin-resistant *Enterobacter* spp. isolates in Korea. *Curr Microbiol.* 2019;76:644–9.
- [6] Lee JY, Ko KS. Mutations and expression of *PmrAB* and *PhoPQ* related with colistin resistance in *Pseudomonas aeruginosa* clinical isolates. *Diagn Microbiol Infect Dis.* 2014;78:271–6.
- [7] Kang KN, Klein DR, Kazi MI, Guérin F, Cattoir V, Brodbelt JS, et al. Colistin heteroresistance in *Enterobacter cloacae* is mediated by *PmrAB*-independent 4-amino-4-deoxy-L-arabinose addition to lipid A. *bioRxiv.* 2019. <http://dx.doi.org/10.1101/516872>
- [8] Miller AK, Brannon MK, Stevens L, Johansen HK, Selgrade SE, Miller SI, et al. *PhoQ* mutations promote lipid A modification and polymyxin resistance of *Pseudomonas aeruginosa* found in colistin-treated cystic fibrosis patients. *Antimicrob Agents Chemother.* 2011;55:5761–9.
- [9] Partridge S, Pilato DV, Doi Y, Feldgarden M, Haft D, Klimke W. Proposal for assignment of allele numbers for mobile colistin resistance (*mcr*) genes. *J Antimicrob Chemother.* 2018;73:2625–30.
- [10] Luo Q, Yu W, Zhou K, Guo L, Shen P, Lu H, et al. Molecular epidemiology and colistin-resistant mechanism of *mcr*-positive and *mcr*-negative clinical isolated *Escherichia coli*. *Front Microbiol.* 2017;8:2262.
- [11] Wang C, Feng Y, Liu L, Wei L, Kang M, Zong Z. Identification of novel mobile colistin resistance gene *mcr-10*. *Emerg Microbes Infect.* 2020;9 (1):508–516.
- [12] The European Committee on Antimicrobial Susceptibility Testing. Breakpoint tables for interpretation of MICs and zone diameters. Version 9.0. 2019. <http://www.eucast.org>
- [13] Filioussis G, Kachrimanidou M, Christodoulou G, Kyritsi M, Hadjichristodoulou C, Adamopoulou M, et al. Bovine mastitis caused by a multidrug-resistant, *mcr-1*-positive (colistin-resistant), extended-spectrum β -lactamase-producing *Escherichia coli* clone on a Greek dairy farm. *J Dairy Sci.* 2019;103 (1):852–857.
- [14] Lei CW, Zhang Y, Wang YT, Wang HN. Detection of mobile colistin resistance gene *mcr-10.1* in a conjugative plasmid from *Enterobacter roggenkampii* of chicken origin in China. *Antimicrob Agents Chemother.* 2020;64:e01191–20.
- [15] Choi Y, Lee JY, Lee H, Park M, Kang K, Lim SK, et al. Comparison of fitness cost and virulence in chromosome- and plasmid-mediated colistin-resistant *Escherichia coli*. *Front Microbiol.* 2020;11:798.

- [16] Jayol A, Nordmann P, Brink A, Poiriel L. Heteroresistance to colistin in *Klebsiella pneumonia* associated with alterations in the PhoPQ regulatory system. *Antimicrob Agents Chemother*. 2015;59:2780–4.
- [17] Gunn JS, Miller SI. PhoP-PhoQ activates transcription of pmrAB, encoding a two-component regulatory system involved in *Salmonella typhimurium* antimicrobial peptide resistance. *J Bacteriol*. 1996;178:6857–64.
- [18] Moskowitz SM, Ernst RK, Miller SI. PmrAB, a two-component regulatory system of *Pseudomonas aeruginosa* that modulates resistance to cationic antimicrobial peptides and addition of aminoarabinose to lipid A. *J Bacteriol*. 2004; 186:575–9.

SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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